AUTOMORPHISM SHEAVES OF THE FIRST-ORDER LIFTING GERBE ARE QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1322-1334 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1336-1495, the complete author proof (158 body lines).

Editorial note. Navigation stubs: 16 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_stacks-stacks-more-morphisms.json (pb=6696, stmt 1322-1334, proof 1336-1495).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-local-lifts-isomorphic: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1211–1226. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-gerbe-of-lifts-fibred: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 964–967. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-gerbe-of-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 875–903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04TP: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CKF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CKB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CN: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CK8: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 061C: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CK5: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CK7: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M0: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M1: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CJY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1495–1495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Remark remark-gerbe-of-lifts assume $\mathcal{X} \subset \mathcal{X}'$ is a first order thickening. Then*

(1) *the automorphism sheaves of objects of the gerbe $p : \mathcal{C} \to W_{spaces,étale}$ constructed in Remark remark-gerbe-of-lifts are abelian, and*
(2) *the sheaf of groups $\mathcal{G}$ constructed in Stacks, Lemma 0CJY is a quasi-coherent $\mathcal{O}_W$-module.*

Proof. We will prove both statements at the same time. Namely, given an object $\xi = (U, U', a, i, x', \alpha)$ we will endow $Aut(\xi)$ with the structure of a quasi-coherent $\mathcal{O}_U$-module on $U_{spaces,étale}$ and we will show that this structure is compatible with pullbacks. This will be sufficient by glueing of sheaves (Sites, Section 04TP) and the construction of $\mathcal{G}$ in the proof of Stacks, Lemma 0CJY as the glueing of the automorphism sheaves $Aut(\xi)$ and the fact that it suffices to check a module is quasi-coherent after going to an étale covering (Properties of Spaces, Lemma 03M0).

We will describe the sheaf $Aut(\xi)$ using the same method as used in the proof of Lemma lemma-etale-local-lifts-isomorphic. Consider the algebraic space

$$I' = U' \times_{x',\mathcal{X}',x'} U'$$

with projections $p' : I' \to U'$ and $q' : I' \to U'$. Over I' there is a universal 2-morphism $\gamma : x' \circ p' \to x' \circ q'$. The identity $x' \to x'$ defines a diagonal morphism

$$\begin{array}{ccccc} U' & & \xrightarrow{\quad\quad} & & I' \\ & \searrow^{\Delta'} & & \swarrow_{p'} \quad \searrow^{q'} & \\ & & U' & & U' \end{array}$$

such that the compositions $U' \to I' \to U'$ and $U' \to I' \to U'$ are the identity morphisms. We will denote the base change of U', I', p', q', Δ' to $\mathcal{X}$ by U, I, p, q, Δ. Since $W' \to \mathcal{X}'$ is smooth, we see that $p' : I' \to U'$ is smooth as a base change.

A section of $Aut(\xi)$ over U is a morphism $\delta' : U' \to I'$ such that $\delta'|_U = \Delta$ and such that $p' \circ \delta' = \mathrm{id}_{U'}$. To be explicit, $(\mathrm{id}_U, q' \circ \delta', (\delta')^*\gamma) : \xi \to \xi$ is a formula for the corresponding automorphism. More generally, if $f : V \to U$ is an étale morphism, then there is a thickening $j : V \to V'$ and an étale morphism $f' : V' \to U'$ whose restriction to V is f and $f^*\xi$ corresponds to $(V, V', a \circ f, j, x' \circ f', f^*\alpha)$, see proof of Lemma lemma-gerbe-of-lifts-fibred. a section of $Aut(\xi)$ over V is a morphism $\delta' : V' \to I'$ such that $\delta'|_V = \Delta \circ f$ and $p' \circ \delta' = f'$[1].

We conclude that $Aut(\xi)$ as a sheaf of sets agrees with the sheaf defined in More on Morphisms of Spaces, Remark 0CK7 for the thickenings $(U \subset U')$ and $(I \subset I')$ over $(U \subset U')$ via $\mathrm{id}_{U'}$ and p'. The diagonal Δ' is a section of this sheaf and by acting on this section using More on Morphisms of Spaces, Lemma 061C we get an isomorphism

$$\mathcal{H}om_{\mathcal{O}_U}(\Delta^*\Omega_{I/U}, \mathcal{C}_{U/U'}) \longrightarrow Aut(\xi) \tag{0.1.1}$$

[1] A formula for the corresponding automorphism is $(\mathrm{id}_V, h', (\delta')^*\gamma)$. Here $h' : V' \to V'$ is the unique (iso)morphism such that $h'|_V = \mathrm{id}_V$ and such that

$$\begin{array}{ccc} V' & \xrightarrow{h'} & V' \\ & \searrow^{q' \circ \delta'} & \downarrow f' \\ & & U' \end{array}$$

commutes. Uniqueness and existence of h' by topological invariance of the étale site, see More on Morphisms of Spaces, Theorem 05ZH. The reader may feel we should instead look at morphisms $\delta'' : V' \to V' \times_{\mathcal{X}'} V'$ with $\delta'' \circ j = \Delta_{V'/\mathcal{X}'}$ and $\mathrm{pr}_1 \circ \delta'' = \mathrm{id}_{V'}$. This would be fine too: as $V' \times_{\mathcal{X}'} V' \to I'$ is étale, the same topological invariance tells us that sending δ'' to $\delta' = (V' \times_{\mathcal{X}'} V' \to I') \circ \delta''$ is a bijection between the two sets of morphisms.

on $U_{spaces,étale}$. There three things left to check

(1) the construction of (0.1.1) commutes with étale localization,
(2) $\mathcal{H}om_{\mathcal{O}_U}(\Delta^*\Omega_{I/U}, \mathcal{C}_{U/U'})$ is a quasi-coherent module on U,
(3) the composition in $Aut(\xi)$ corresponds to addition of sections in this quasi-coherent module.

We will check these in order.

To see (1) we have to show that if $f : V \to U$ is étale, then (0.1.1) constructed using ξ over U, restricts to the map (0.1.1)

$$\mathcal{H}om_{\mathcal{O}_V}(\Delta_V^*\Omega_{V\times_{\mathcal{X}}V/V}, \mathcal{C}_{V/V'}) \to Aut(\xi|_V)$$

constructed using $\xi|_V$ over V on $V_{spaces,étale}$. This follows from the discussion in the footnote above and More on Morphisms of Spaces, Lemma 0CK8.

Proof of (2). Since p' is smooth, the morphism $I \to U$ is smooth, and hence the relative module of differentials $\Omega_{I/U}$ is finite locally free (More on Morphisms of Spaces, Lemma 0CK5). On the other hand, $\mathcal{C}_{U/U'}$ is quasi-coherent (More on Morphisms of Spaces, Definition 04CN). By Properties of Spaces, Lemma 03M1 we conclude.

Proof of (3). There exists a morphism $c' : I' \times_{p',U',q'} I' \to I'$ such that (U', I', p', q', c') is a groupoid in algebraic spaces with identity Δ'. See Algebraic Stacks, Lemma 04T4 for example. Composition in $Aut(\xi)$ is induced by the morphism c' as follows. Suppose we have two morphisms

$$\delta'_1, \delta'_2 : U' \longrightarrow I'$$

corresponding to sections of $Aut(\xi)$ over U as above, in other words, we have $\delta'_i|U = \Delta_U$ and $p' \circ \delta'_i = \mathrm{id}_{U'}$. Then the composition in $Aut(\xi)$ is

$$\delta'_1 \circ \delta'_2 = c'(\delta'_1 \circ q' \circ \delta'_2, \delta'_2)$$

We omit the detailed verification[2]. Thus we are in the situation described in More on Groupoids in Spaces, Section 0CKB and the desired result follows from More on Groupoids in Spaces, Lemma 0CKF. □

[2]The reader can see immediately that it is necessary to precompose δ'_1 by $q' \circ \delta'_2$ to get a well defined U'-valued point of the fibre product $I' \times_{p',U',q'} I'$.